

PROCUREX

National e-Procurement Mandi Management Platform & Real-Time Queue Ecosystem

Target Problem: SIH 2026 PS 26032	Domain: Agri-Tech & Rural Governance	Live URL: kissan-queue.onrender.com
Core Stack: React 18, Node.js, MongoDB, Socket.IO	Languages: English, Tamil (■■■■■■), Hindi (■■■■■■■)	Channels: Web + 2G SMS / USSD Offline

1. Executive Pitch & Value Proposition (The Hook)

The Ground Reality: Across India's 7,000+ Agricultural Produce Market Committees (APMCs/Mandis), millions of farmers face grueling **18 to 48-hour physical waiting lines** in tractor queues during peak harvest seasons. Overcrowding leads to grain spoilage, distress selling to exploitative middlemen, traffic gridlock, and administrative delays in MSP calculation.

The ProcureX Solution: ProcureX is a production-grade, government-certified digital procurement platform that completely eliminates physical waiting queues through **atomic slot scheduling, live digital token broadcasting, electronic weighbridge integration, automated state grain logistics, and 2G feature phone access.**

■ 0 Hours Waiting: Pre-booked 2-hr slots prevent Mandi road congestion.	■■ Fraud-Proof MSP: Auto-computed ■ payout based on net scale & moisture.
■ Automated Logistics: Weighment auto-dispatches trucks to state silos.	■ 100% Inclusive: Full SMS and USSD (*999*26032#) for non-smartphone users.

2. Dedicated Role Portals & End-to-End Workflow

Portal Role	Target Users	Core Capabilities & Operational Responsibilities
■■■ Farmer Portal	Farmers, FPOs, Village Cooperatives	3-Step Slot Booking (Crop, Mandi, 2-Hr Window) - Live WebSocket Queue Position Tracker & SMS Tokens - QR-Encoded Official Digital e-Slips & Aadhaar DBT Status
■ Mandi Authority	Mandi Secretaries, Yard Staff, Weighbridge Operators	- Gate Check-In & Live Call Next Farmer Desk - Electronic Weighbridge Net Weight & Moisture % Logging - Automatic MSP Calculation & Digital e-Slip Issuance - Dynamic Slot Schedule & No-Show Reallocation
■ Logistics Fleet	State Grain Logistics Cell, Truck Drivers	- Auto-Created Grain Transport Tasks from Mandi Weighments - Vehicle (HR-05-CD-9988) & Driver Assignment Modal - Real-time Milestone Tracking (Assigned → In Transit → Delivered)
■■ State Admin	Ministry Officials, State Agri Directors	- State-wide Procurement Analytics & Crop Distribution Charts - APMC Mandi Oversight, Capacity Utilization, & Efficiency Ratios - User Management & Cryptographically Logged Audit Trail

3. Winning Pitching Strategy (The 5-Minute Master Deck)

Deliver this structured narrative to win hackathons, panel evaluations, and government stakeholder presentations:

Time	Pitch Section	Key Narrative & What to Say to Judges
0:00 - 0:45	1. The Hook & Ground Problem	<i>'Imagine traveling 40 km with a tractor full of wheat, only to sleep on the road for 2 days in a 2-kilometer Mandi line.'</i> State that this causes ₹5,000 Cr in post-harvest losses annually in India.
0:45 - 1:30	2. The ProcureX Solution	Introduce ProcureX as the unified digital ecosystem that synchronizes farmer slots, gate queues, weighbridge scale sensors, and state transport silos.
1:30 - 3:30	3. Live Connected Tech Demo	Execute the real 4-role loop live: Farmer books slot → Mandi calls token → Weighs produce & issues slip → Logistics automatically gets transport task & assigns fleet. (Shows real data flow!)
3:30 - 4:15	4. Technical Depth & Inclusivity	Highlight atomic concurrency (no double booking), 2G SMS/USSD simulator for feature phones, Socket.IO live sync, and 100% multilingual support (English, Tamil, Hindi).
4:15 - 5:00	5. Impact, Scale & Q&A; Hook	Conclude with scalability metrics: 100% cloud deployment on Render + MongoDB Atlas, direct Aadhaar DBT integration readiness, and seamless APMC adoption.

4. Pro-Tips for Hackathons & Jury Defense (Q&A; Secrets)

- **Tip 1: Show, Don't Just Tell:** Do NOT stay on PowerPoint slides for more than 90 seconds. Jump straight into the live interactive application.
- **Tip 2: Demonstrate the Non-Smartphone Story:** Open the built-in **SMS / USSD Simulator (*999*26032#)** modal to prove you thought of illiterate and feature phone farmers.
- **Tip 3: Language Switching as a Wow-Factor:** Switch between English, Tamil, and Hindi live in front of the judges using the top-right header language dropdown.
- **Tip 4: Highlight Zero Hardcoded Data:** Open MongoDB / Audit Logs to prove every token, procurement receipt, and truck dispatch is stored in real databases.
- **Tip 5: Explain No-Show Slot Reallocation:** When asked *'What if a farmer books and does not show up?'* explain that after 3 calls, the slot is automatically released to waitlisted farmers.
- **Tip 6: Concurrency Defense:** When asked *'What if 100 farmers book the same slot at 9:00 AM?'* state that MongoDB atomic `findOneAndUpdate` with remaining capacity checks prevents overbooking.
- **Tip 7: Emphasize Automatic Logistics Handover:** Emphasize that when Mandi completes weighment, the Logistics dashboard automatically updates via WebSockets without manual phone calls.
- **Tip 8: Clear Role Separation:** Use 4 different browser tabs (or incognito) to show real-time synchronization between the Farmer, Mandi Desk, and Logistics Fleet.
- **Tip 9: Print e-Slip Receipt Live:** Click 'Print e-Receipt' on a completed procurement to showcase the QR-authenticated Government of India voucher.
- **Tip 10: State Administrator Oversight:** Show the Recharts crop distribution graph and immutable audit log to demonstrate government compliance.

5. Live Demo Cheat Sheet & Verified Test Credentials

Use these pre-configured verified credentials during live presentation or testing:

Role	Name	Login Mobile / Email	Password	Target Route
■■■ Farmer	Sardar Gurpreet Singh	9876500001 farmer@procurex.gov.in	Password@123	/farmer
■ Mandi Desk	Shri Om Prakash	9999900002 authority@procurex.gov.in	Password@123	/storage
■ Logistics Fleet	Food Grain Fleet Cell	9999900004 logistics@procurex.gov.in	Password@123	/logistics
■■ State Admin	Dr. Rajiv Malhotra (IAS)	9999900001 admin@procurex.gov.in	Admin@123	/admin

6. Architectural Innovation & National Scalability

Atomic Concurrency Engine: MongoDB findOneAndUpdate with condition checks guarantees zero double-booking even under peak seasonal harvest spikes.	Bidirectional WebSockets: Instant socket room broadcasts trigger turn alerts to farmers and new transport task alerts to logistics.
Direct Benefit Transfer (DBT): Automatic ■ MSP calculation based on gross scale minus tare weight with moisture grading deductions.	State-Wide Scalability: Microservice-ready REST APIs ready for integration with national e-NAM (National Agriculture Market) and FCI Silos.

ProcureX is built with pride for Smart India Hackathon 2026 Problem Statement 26032 to transform agricultural procurement across India. Dedicated to empowering our farmers (Jai Jawan, Jai Kisan).